MIL-E-1/1147 A (Sig C)
17 July 1959
Superseding
MIL-E-1/1147(SigC)
30 September 1958

INDIVIDUAL MILITARY SPECIFICATION SHEET
ELECTRON TUBE, MICROWAVE NOISE SOURCE
TYPE 6782

Applicable paragraphs of the latest issue of Military Specification MIL-E-1, as referenced herein, form a part of this specification sheet.

Description: Gaseous Discharge Diode, S Band (Note 12)

Ratings:

	If	Ib	Epp	TA	T Bulb
	ma	mAdc	Vac	°C	°C
Absolute Maximum	--	--	--	+85	+125
Minimum	--	--	2000	-55	--
Test Conditions	0	250	--	--	--

Cathode: Filamentary Type.
Dimensions: Per Outline Drawing. (Fig. 3)
Base: Miniature Bi-pin, per Outline Drawing (Fig. 3)
Mounting Position: Any.

Ref. Para	Test	Conditions	Sym.	Min.	Max.	Units
-------	Qualification:	Required. Notes 13-14				
4.5	Holding Period:	168 hours.				
4.9.18.1.8	Carton Drop:					
4.9.20.3	*Vibration:	(No Voltages) Note 10				
4.10.5.1	Filament Voltage:	If-1 = 170 mAdc	Ef-1	--	15.0	Vdc
		If-2 = 170 mAdc	Ef-2	--	15.0	Vdc
4.13.2	Tube Voltage Drop:	Forward & Reverse Notes 1,2	Etd-a	41	48	Vdc
			Etd-b	41	48	Vdc
4.13.1	Ionization Voltage:	Rp = 8000 ohms Note 3.	Ez	---	2000	Vac
------	Excess Noise Ratio:	F = 3300 Mc. Notes 3, 4, 6, 7, 11	Nr -1	8.60	9.10	db
------	* Match (1)	F = 3270 Mc. Notes 4,6. I_b = 250 mAdc	VSWR	1.30:1	1.38:1	--
------	* Match (2)	F = 3270 Mc. Notes 4,6. Ib= 0 mAdc	VSWR	---	1.02:1	--

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Ref. Ref. Para.	Test	Conditions	Sym.	Min.	Max.	Units
-----	Intermittent Life Test:	Epp= 2000 Vac. Rp = 8000 ohms Notes 3,8,9. (One min. on, Two min. off)	No.	2500	---	Cycles
4.11.4	Intermittent Life Test End Points:	Excess Noise Ratio: Note 11.	Nr-1	8.55	9.15	db
		Ionization Voltage:	Ez	---	2000	Vac

Note 1. The tube shall be tested in the circuit of Fig. 1.

Note 2. In the test circuit of Fig. 1, with a filament current of 170 mAdc, the tube shall operate within three tries. The tube shall be tested and then reverse tested using alternate ends as anode and cathode. One reading of Etd shall be taken not less than 15 seconds after ionization, in each direction.

Note 3. The tube shall be tested in total darkness.

Note 4. The test shall be made with the tube mounted in the Noise Tube Mount (Fig. 4) of this specification. Termination shall be made by a matched RG-48/U termination, having a minimum VSWR of 1.01:1 such as Hewlett-Packard S-914A, or equivalent.

Note 5. The frequency specified is that of the local oscillator.

Note 6. The frequency specified is that of the signal generator.

Note 7. Excess Noise Ratio shall be measured by comparison with an approved standard. The Excess Noise Ratio measurement tests shall be made using the circuit of block diagram Fig. 2 or equivalent. The VSWR, looking into the balanced mixer shall not be greater than 2:1. The excess noise ratio measurement tests shall be made, using the circuit of block design Fig. 2 or equivalent. The VSWR, looking into the balanced mixer shall not be greater than 2:1.

Note 8. The tube shall be tested at an ambient temperature of + 85°C.

Note 9. Life test samples shall consist of one tube per lot. A lot shall consist of one weeks production. No lot shall be shipped until the completion of life test. Failure of initial samples shall require a successful life test of two additional tubes before release of a lot for shipment.

Note 10. Intermittent life test end points shall apply.

Note 11. The excess Noise Ratio (Nr -1) is defined in db, as $Nr - 1 = 10 \log \left(\frac{Te}{290} - 1\right)$ where Te is the effective electron temperature.

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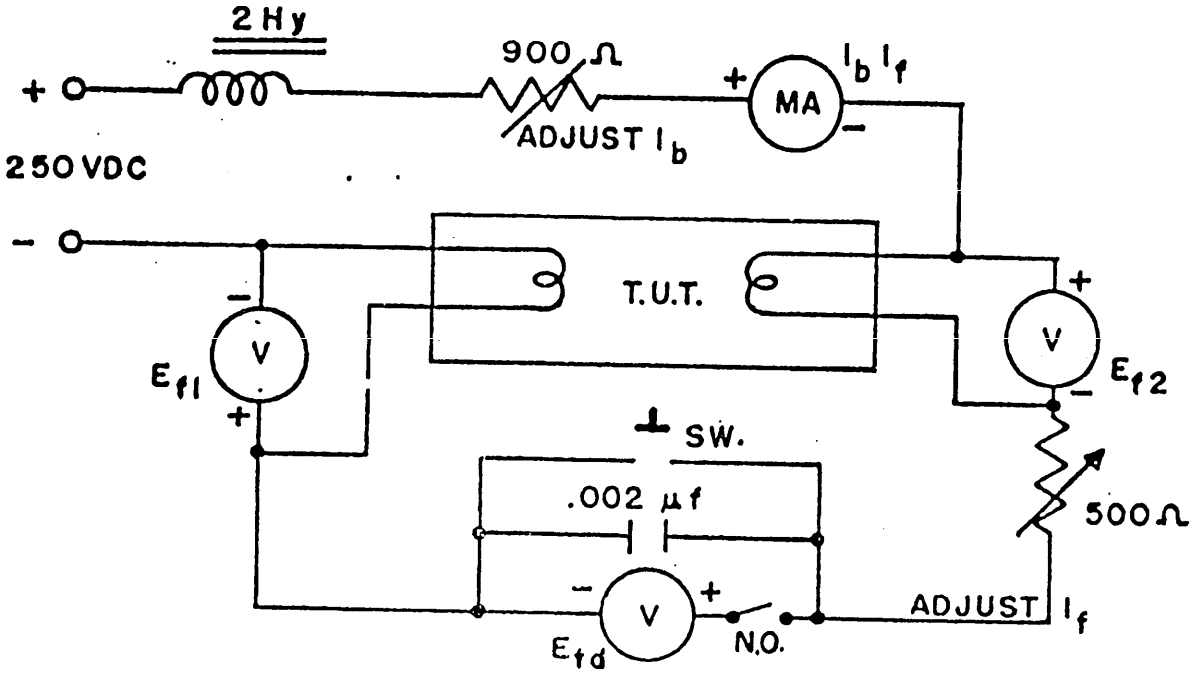
Note 12. The noise frequencies generated by this tube cover a broad band of frequencies. This bandwidth is limited only by the type of Mount used. This tube is normally used with a mount in RG-48/U wave guide, at a 90 degree angle in the H plane. Other wave guide sizes may be used with properly adapted mounts.

Note 13. Application for Qualification Inspection should be made in accordance with Appendix A of Military Specification MIL-E-1, with the exception that the request for Qualification Inspection shall be addressed to Components and Materials Branch, Standardization Division, U. S. Army Signal Equipment Support Agency, Fort Monmouth, New Jersey.

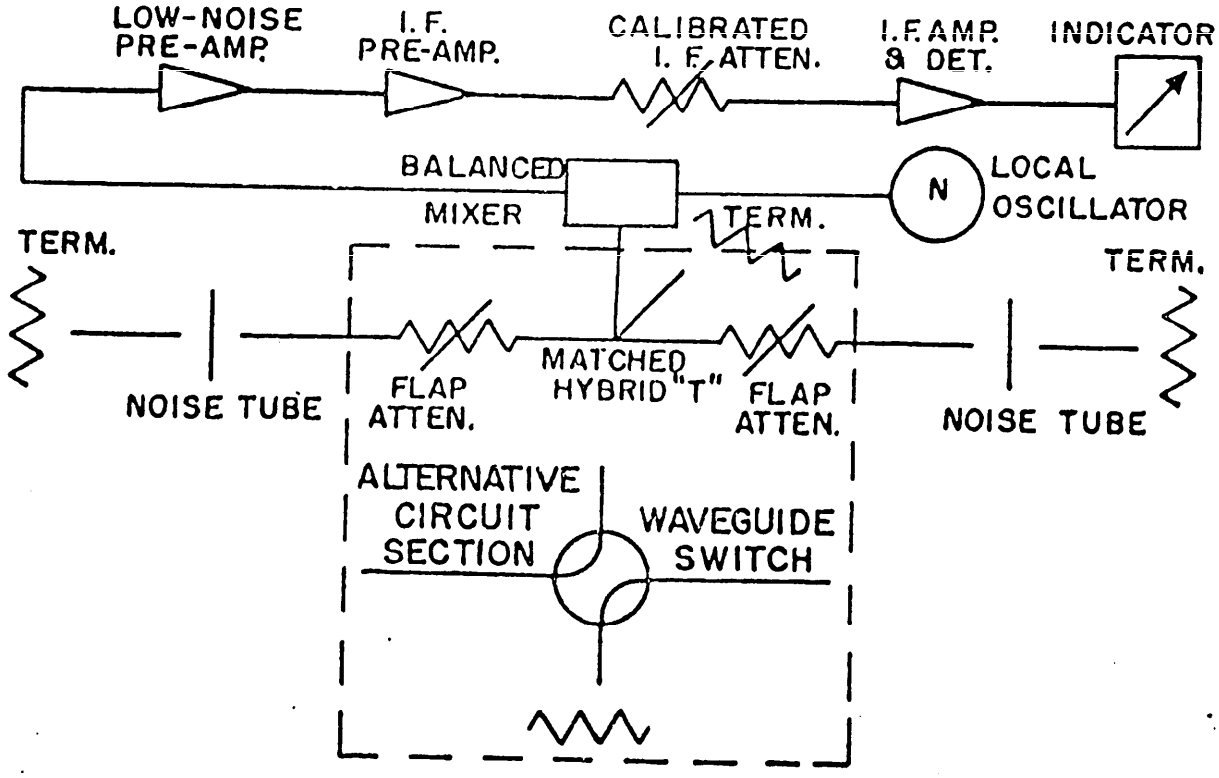
Note 14. Electron Tube type 6782 procured directly or indirectly on Army Contracts, shall be marked "US Army" either above or preceding the type designation as specified in the bid request or contract. The letters "JAN" or any abbreviation thereof shall not be used.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

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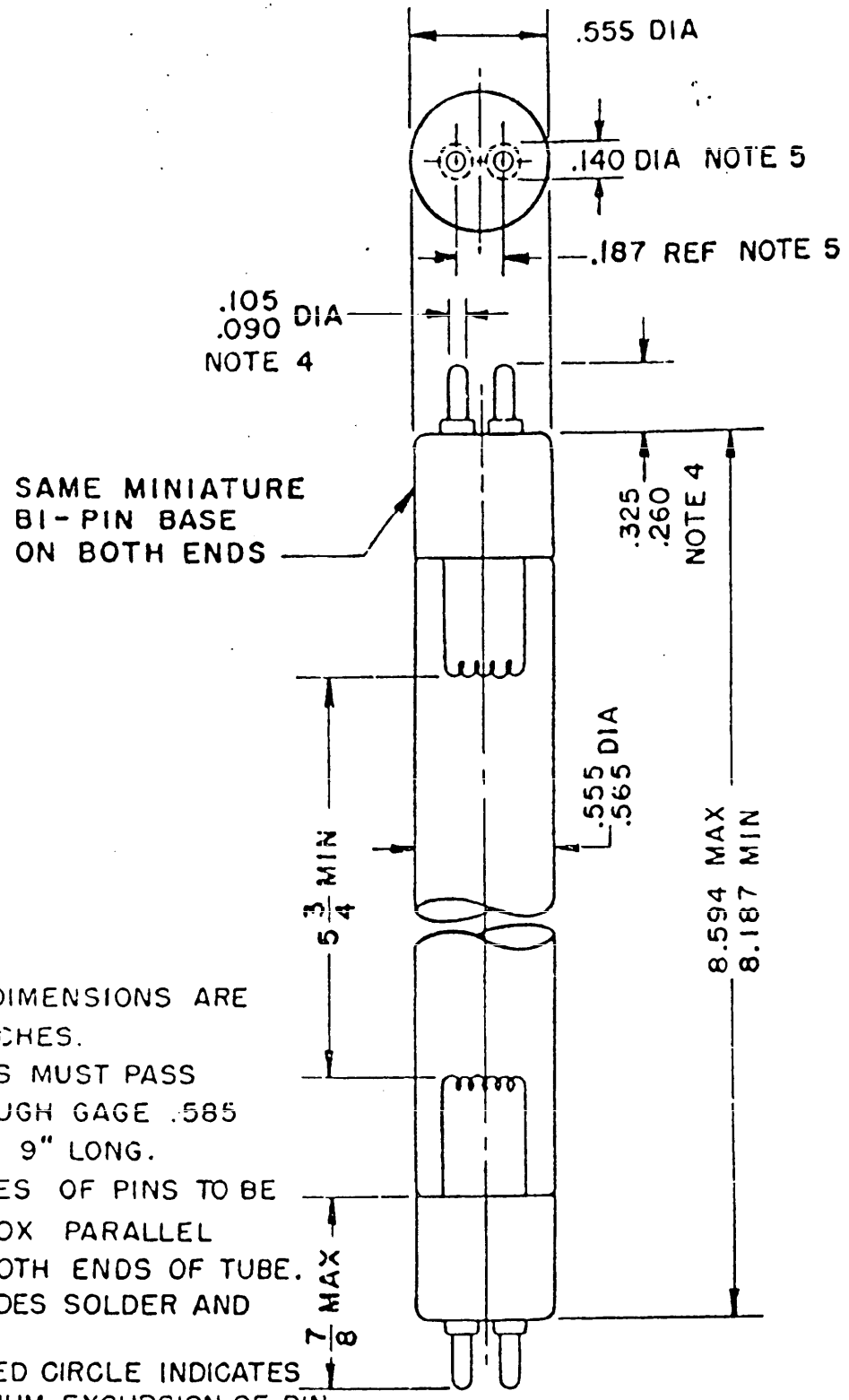


D. C. TEST CIRCUIT
FIG. 1



TEST CIRCUIT FOR EXCESS NOISE MEASUREMENTS
FIG. 2

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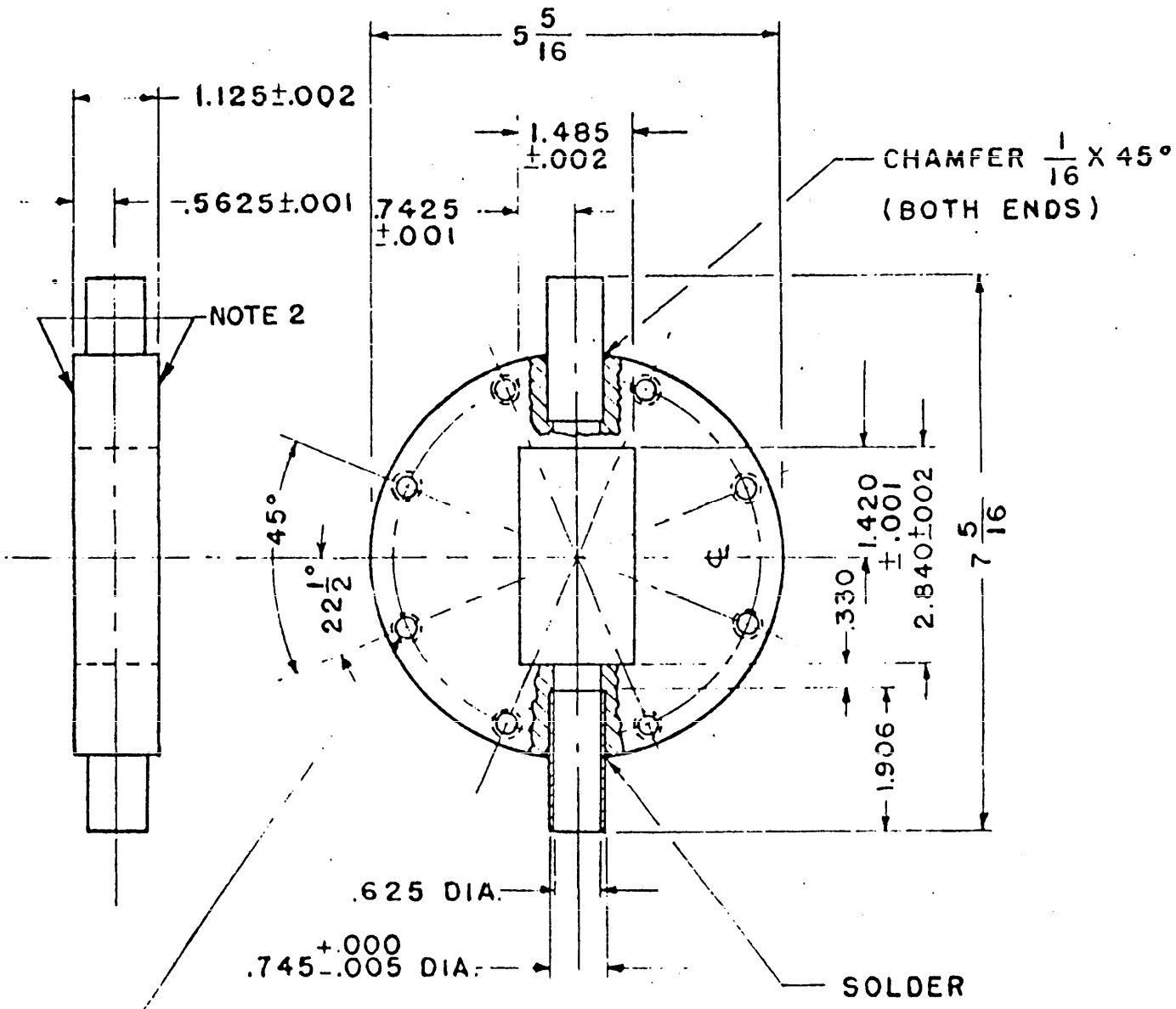
NOTE:

1- ALL DIMENSIONS ARE IN INCHES.
2- TUBES MUST PASS THROUGH GAGE .585 DIA x 9" LONG.
3- PLANES OF PINS TO BE APPROX PARALLEL ON BOTH ENDS OF TUBE.
4- INCLUDES SOLDER AND WIRE.
5- DOTTED CIRCLE INDICATES MAXIMUM EXCURSION OF PIN INCLUDING VARIATIONS IN CENTER TO CENTER SPACING, PIN DIAMETER AND TILT. REFERENCE DIMENSION IS CENTER TO CENTER OF .140" DIA CIRCLES.

ELECTRON TUBE, TYPE 6782 (NOISE SOURCE)

FIG. 3

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NOTE:

1 - ALL DIMENSIONS ARE IN INCHES.

2 - SIDES SHALL BE PARALLEL WITHIN .002 TOTAL INDICATOR READING.

MOUNT FOR ELECTRON TUBE, TYPE 6782 (NOISE SOURCE)

FIG. 4

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